Mathematical Statistics

Lecture 12: The Poisson Distribution

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November 19, 2025

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1. Introduction: From Binomial to Poisson

The **Binomial Distribution** is a powerful tool for modeling the number of successes in a fixed number of independent trials (n), where each trial has the same probability of success (p). To use it, we must know n, p, and x.

1.1. The Real-World Challenge

In many real-life scenarios, counting the total number of trials (n) is impossible or impractical. Consider a YouTube video where we are told that, on average, it receives **2 likes per day**. We don't know the number of daily views (n) or the exact probability of a single viewer leaving a like (p). This is where the Poisson distribution is essential.

When to use the Poisson Distribution

The Poisson distribution is used to model the number of times an event occurs within a specified interval of time or space when you only know the **average rate of occurrence**. This average rate is denoted by the Greek letter **lambda** (λ). Examples include:

- Likes per day ($\lambda = 2$).
- Accidents per month on a highway.
- Patients arriving at an emergency room per hour.

2. The Mathematical Foundation of Poisson

The Poisson distribution is a **limiting case of the Binomial distribution**. This connection explains its underlying assumptions. To model an average rate (λ) using a Binomial approach, we would have to make the time intervals for each trial infinitesimally small to ensure independence.

This leads to two crucial insights:

1. As the time interval for each trial becomes infinitesimally small, the number of trials (n) in a fixed period (like one day) must approach **infinity** ($n \to \infty$).

2. The mean of the Binomial is np, which must equal our finite average rate λ. If $n \to \infty$ and $np = \lambda$, then the probability of success in any one trial (p) must approach **zero** ($p \to 0$).

Conditions for Poisson Distribution

The Poisson distribution models events that happen when:

1. The number of possible trials is very large ($n \to \infty$).
2. The probability of success in any given trial is very small ($p \to 0$).

This makes it perfect for modeling **rare events**.

3. Properties of the Poisson Distribution

3.1. Probability Mass Function (PMF)

Definition: Poisson PMF

The probability of observing exactly x events in an interval, given an average rate λ, is:

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

where $x = 0, 1, 2, \dots$ and $e \approx 2.71828$.

3.2. Mean and Variance

One of the most remarkable properties of the Poisson distribution is that its mean and variance are equal to the rate parameter λ.

- **Mean (Expected Value):** $\mu = E[X] = \lambda$
- **Variance:** $\sigma^2 = \mathrm{Var}(X) = \lambda$

Derivation Sketch: The mean is found by $E[X] = \sum x \cdot P(X = x)$, which simplifies to λ after recognizing the Taylor series for e^λ. The variance is found using $\mathrm{Var}(X) = E[X^2] - (E[X])^2$. Calculating $E[X^2]$ shows it is equal to $\lambda^2 + \lambda$. Therefore,

$$\mathrm{Var}(X) = (\lambda^2 + \lambda) - \lambda^2 = \lambda$$

4. Real-World Applications and Solved Examples

Example Highway Accidents

Problem: The average number of accidents on a national highway is **1.8 per day** ($\lambda = 1.8$). Determine the probability that the number of accidents is (a) at least one, and (b) at most one.

Solution:

a) **Probability of at least one accident:** $P(X \geq 1)$
This is calculated as the complement: $1 - P(X = 0)$.

$$P(X = 0) = \frac{e^{-1.8}(1.8)^0}{0!} = e^{-1.8} \approx 0.1653$$

Therefore, $P(X \geq 1) = 1 - 0.1653 = 0.8347$. There is an 83.5% chance of at least one accident.

b) **Probability of at most one accident:** $P(X \leq 1)$
This is $P(X = 0) + P(X = 1)$.

$$P(X = 1) = \frac{e^{-1.8}(1.8)^1}{1!} = e^{-1.8} \cdot 1.8 \approx 0.2975$$

Therefore, $P(X \leq 1) = 0.1653 + 0.2975 = 0.4628$. There is a 46.3% chance of having zero or one accident.

Example Seed Germination (Poisson Approximation to Binomial)

Problem: A company sells seeds in packets of 200. On average, 5% fail to germinate. The company guarantees 90% germination. What is the probability that a packet violates this guarantee?
Analysis: This is a Binomial problem with $n = 200, p = 0.05$. Since n is large and p is small, we can approximate with Poisson. The average number of failures per packet is $\lambda = np = 200 \times 0.05 = 10$.
Solution:

- The guarantee is 90% germination, so at most 10% (or $0.10 \times 200 = 20$ seeds) can fail.
- A packet *violates* the guarantee if **more than 20 seeds** fail ($X > 20$).
- We calculate $P(X > 20) = 1 - P(X \leq 20)$.
- Calculating the cumulative sum $P(X \leq 20) = \sum_{x=0}^{20} \frac{e^{-10}10^x}{x!}$ using statistical software yields ≈ 0.9984.
- The probability of violating the guarantee is:

$$P(X > 20) = 1 - 0.9984 = 0.0016$$

There is only a 0.16% chance a packet will fail the guarantee.

5. Summary & Key Takeaways

- **Use Case:** Apply Poisson for counting events over a continuous interval (time, area, etc.) when you only know the **average rate** (λ).
- **Connection to Binomial:** Poisson is the limit of the Binomial distribution for large n and small p.
- **PMF Formula:** $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$.
- **Key Property:** The mean and variance are both equal to λ.
- **Applications:** Widely used in quality control, operations research, biology, and finance to model phenomena like customer arrivals, defects, or support calls.

6. Practice Problems

1. **Emergency Room Arrivals:** A hospital emergency room gets an average of 5 patients per hour. What is the probability that they will get exactly 3 patients in the next hour?
2. **Website Traffic:** A popular blog receives an average of 20 comments per day. What is the probability that it will receive no comments tomorrow?

3. **Quality Control:** A factory produces fabric with an average of 0.5 defects per 10 square meters. What is the probability that a 20 square meter piece of fabric will have at least one defect? (Hint: Adjust λ for the new area).

4. **Rare Disease:** A rare disease occurs in 1 out of every 1000 people. In a city of 5000 people, what is the probability that exactly 6 people have the disease? (Use the Poisson approximation to the Binomial).